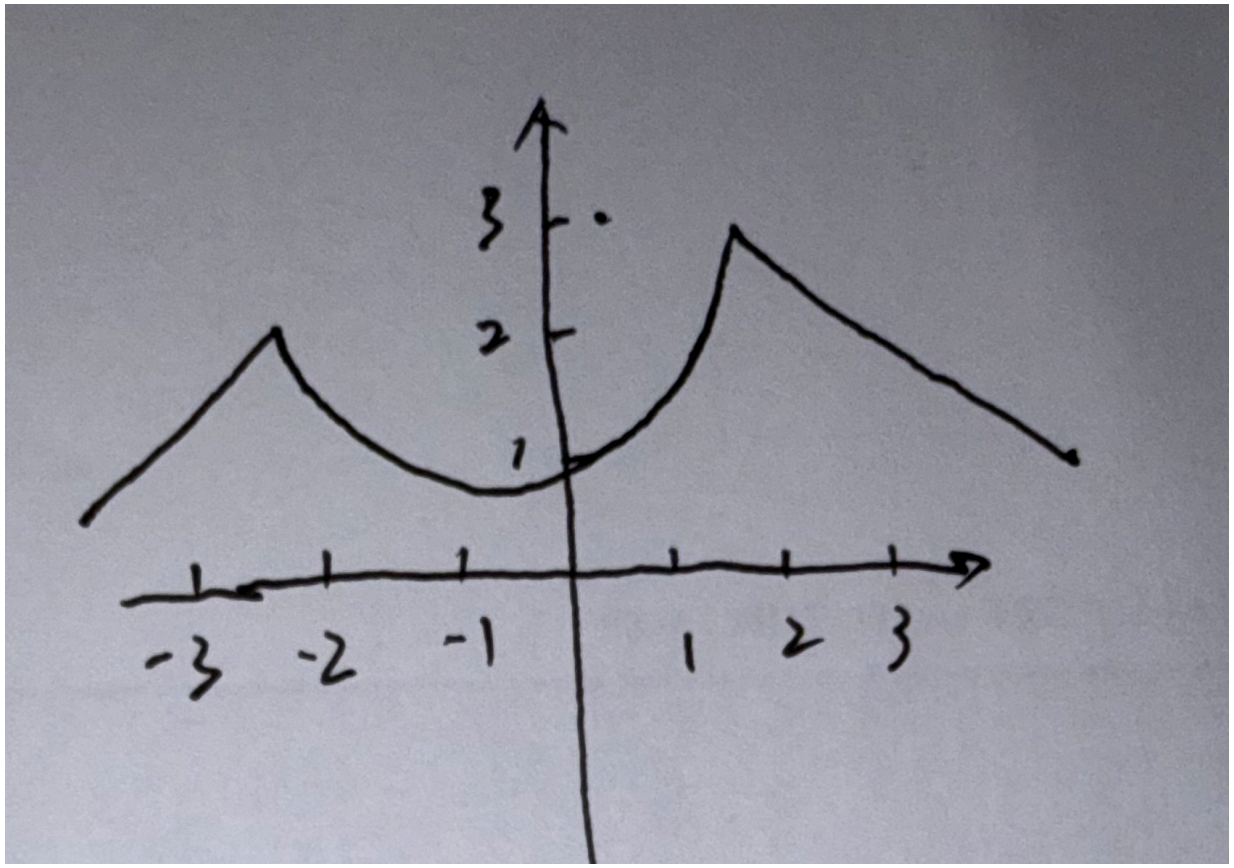
Student ink in digital assessment

A photograph can preserve the work a student actually drew.

Re-entering it as an ideal equation can erase corners, gaps and mathematical mistakes.

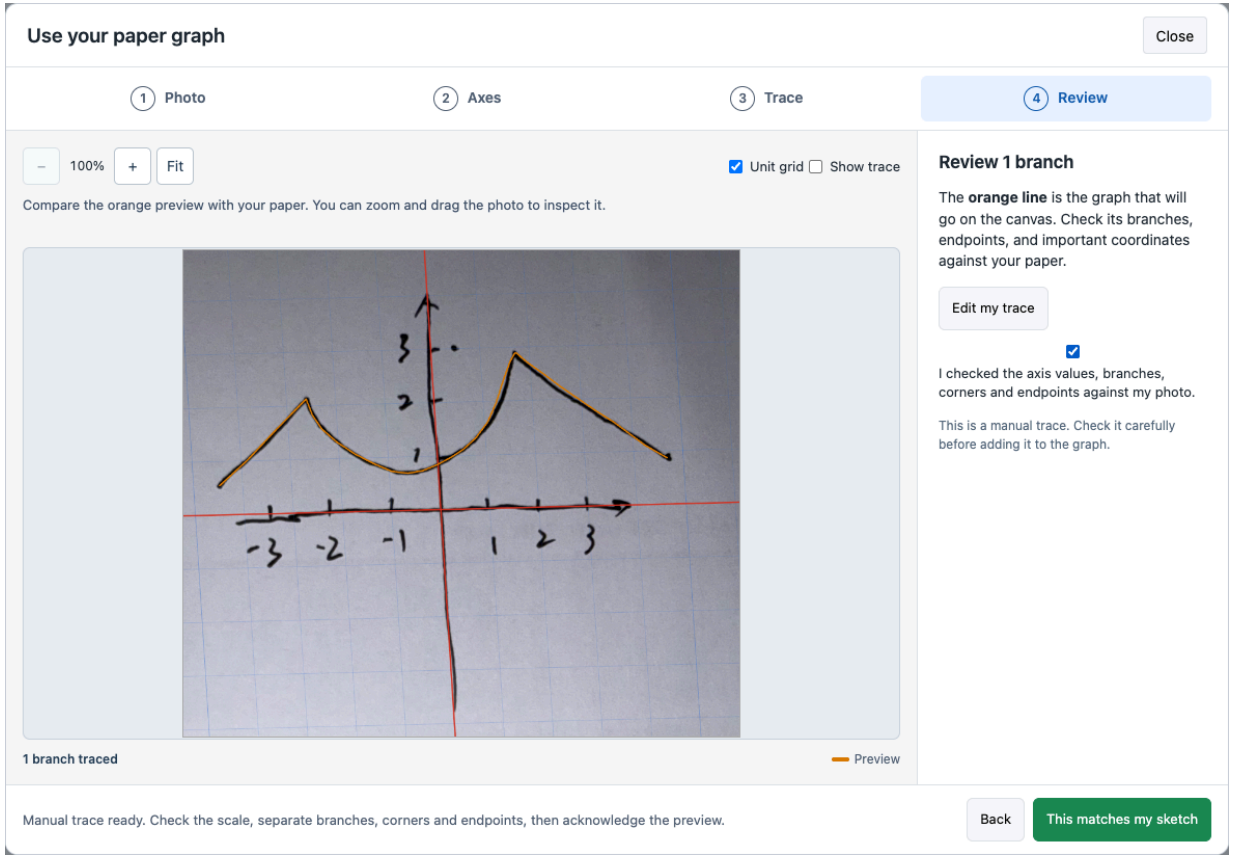
STAR photo-to-PrairieLearn demonstration
November 9, 2026



New user-drawn paper photograph. Detail view; the package includes the full original pixel image.

The assisted workflow

1. Upload the photograph
2. Place axes and enter reference values
3. Trace the actual curve
4. Review, acknowledge and import
5. Edit, Save & Grade, then refresh



Manual review on the new photograph. No equation fitting.

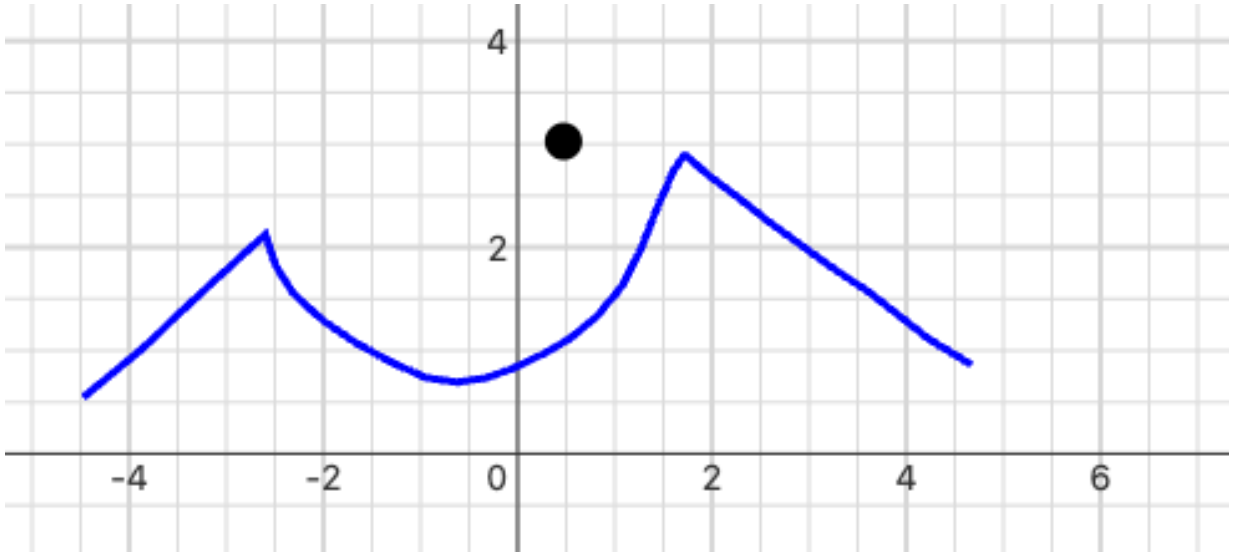
A real-photo assisted demonstration

Both peaks and the off-origin minimum remain visible.

The separate dot is placed manually with **Other marks**.

Prepared native variant
Fresh live attempt

The 0% sandbox grade preserves an incorrect answer.



Native canvas detail. Full canvas and existing test object are in the backup.

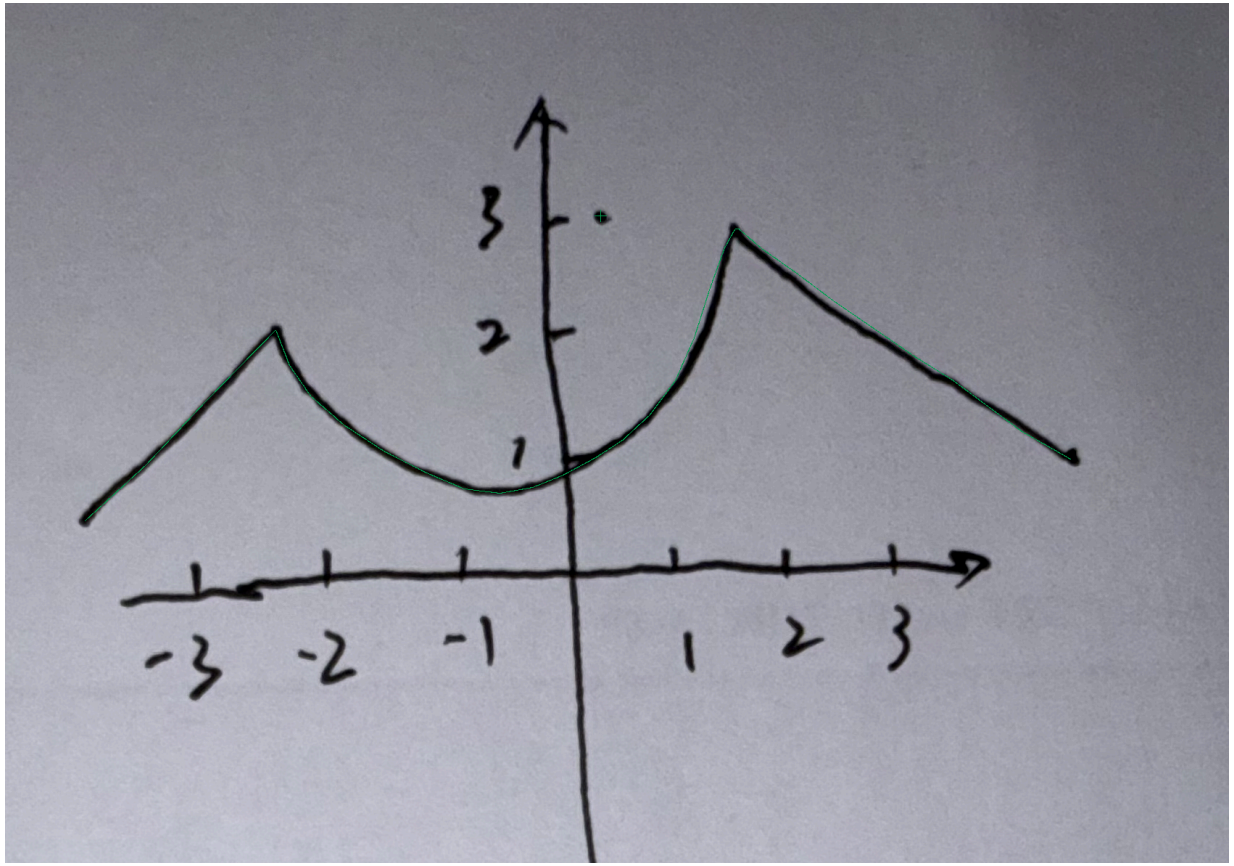
Measured geometry preservation

3.50 px

Bidirectional p95 distance to an approximate visual centerline on this new photo.

Both marked peaks within **2.85 px**. Added rendering drift below **0.001 source px**.

One new paper drawing. Single-observer reference, about 6 px uncertainty. These are geometry diagnostics, not general recognition accuracy.



Green reconstruction on the source photo; cross marks the separately placed dot.

Automatic and manual responsibilities

Person	System
Axis placement and values	Coordinate mapping
Curve tracing	Guarded native import
Dot placement	Editable native objects
Final review	Submission and restoration

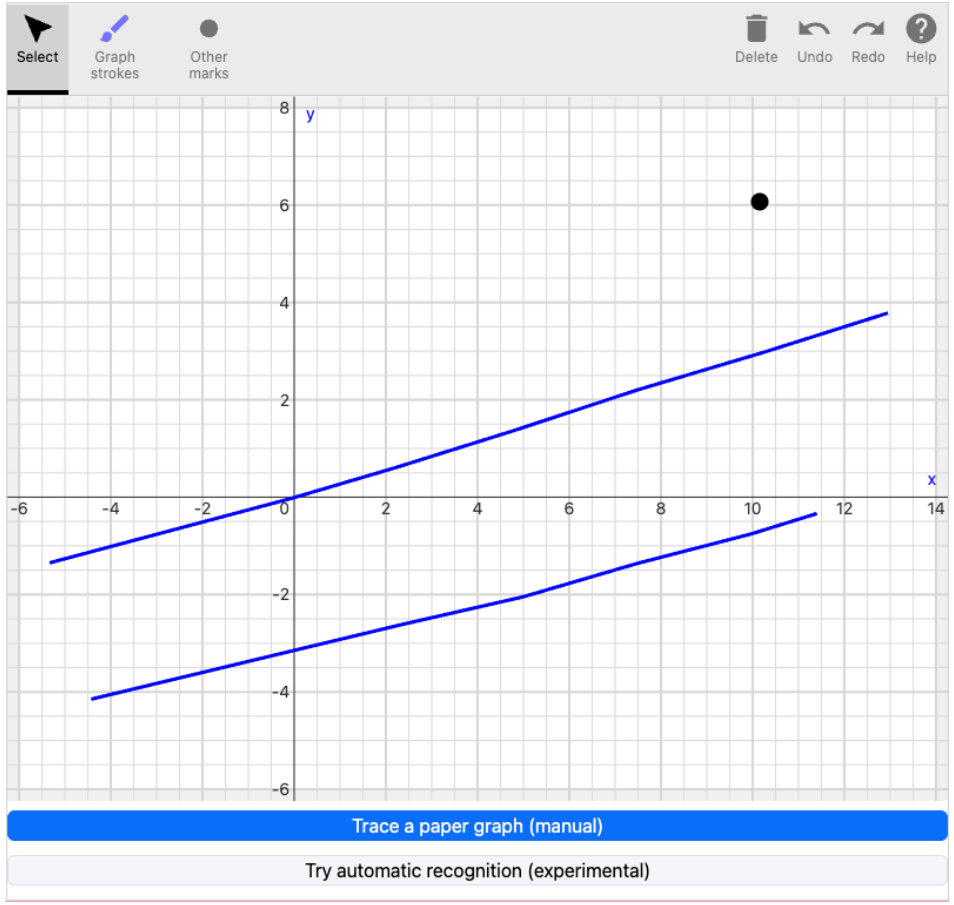
This demonstration uses no automatic recognition, equation fitting or VLM inference.

Reliability and future research

Recorded offline backup works independently of PrairieLearn.

Gemini recognition remains separate research, with permission, quota and $0-cost gates.

Current acceptance is conditional on the rehearsed Mac/Chrome route. Safari automation and physical projector readability need the manual checklist.



Existing two-branch case: separate native objects survived editing and restoration.